# 第二节　求导法则及高阶导数

- 函数的和、差、积、商的导数
- 反函数的求导法则
- 复合函数的求导法则
- 高阶导数

## 求导法则

若$f(x), g(x)$在点$x$可导，则$f(x)$与$g(x)$的和、差、积、商(分母不为零时)在$x$处可导，且

1. $(f(x) \pm g(x))' = f'(x) \pm g'(x)$.
2. $(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$.
3. $\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)} \quad (g(x) \neq 0)$.

# 导数四则运算法则的证明–加减法

$$\left(f(x)+g(x)\right)'=\lim_{\Delta x\to 0}\frac{\left(f(x+\Delta x)+g(x+\Delta x)\right)-\left(f(x)+g(x)\right)}{\Delta x}$$

$$=\lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x}+\frac{g(x+\Delta x)-g(x)}{\Delta x}$$

$$=f'(x)+g'(x)$$

同理 $\left(f(x)-g(x)\right)'=f'(x)-g'(x)$

# 导数四则运算法则的证明–乘法

$$\left(f(x)g(x)\right)' = \lim_{\Delta x \to 0} \frac{f(x+\Delta x)\cdot g(x+\Delta x) - f(x)g(x)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \left( \frac{f(x+\Delta x)-f(x)}{\Delta x} \cdot g(x+\Delta x) + f(x) \frac{g(x+\Delta x)-g(x)}{\Delta x} \right)$$

可导⇒连续

$$= f'(x)g(x) + f(x)g'(x)$$

# 导数四则运算法则的证明–除法

$$\left(\frac{f(x)}{g(x)}\right)' = \lim_{\Delta x \to 0} \frac{\frac{f(x+\Delta x)}{g(x+\Delta x)} - \frac{f(x)}{g(x)}}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x)\cdot g(x) - f(x) g(x+\Delta x)}{\Delta x \cdot g(x+\Delta x) g(x)}$$

$$= \lim_{\Delta x \to 0} \frac{\frac{f(x+\Delta x)-f(x)}{\Delta x} g(x) - f(x) \frac{g(x+\Delta x)-g(x)}{\Delta x}}{g(x+\Delta x)\cdot g(x)}$$

$$= \frac{f'(x) g(x) - f(x) g'(x)}{g^2(x)}$$

## 一些推论

若下面所列函数在点$x$处都可导，则

1. $(f_1(x)+f_2(x)+\cdots+f_n(x))'$
$= f_1'(x)+f_2'(x)+\cdots+f_n'(x)$

2. $(cf(x))' = c'\cdot f(x)+cf'(x) = cf'(x)$

3. $(f(x)g(x)h(x))' = f'(x)(g(x)\cdot h(x))+f(x)\cdot(g(x)h(x))'$
$= f'(x)g(x)h(x)+f(x)g'(x)h(x)+f(x)g(x)h'(x)$

4. $\left(\frac{1}{f(x)}\right)' = \frac{-f'(x)}{f^2(x)}$ eg: $\left(\frac{1}{x}\right)' = -\frac{1}{x^2}$

# 例题

例1 设$y=\sqrt{x}(x^3-4\cos x-\sin 1)$，求$y'$及$y'|_{x=1}$。

解：$y'=(\sqrt{x})'(x^3-4\cos x-\sin 1)+\sqrt{x}\cdot(x^3-4\cos x-\sin 1)'$

$=\frac{1}{2\sqrt{x}}(x^3-4\cos x-\sin 1)+\sqrt{x}(3x^2+4\sin x)$

$y'|_{x=1}=\frac{7}{2}-2\cos 1+\frac{7}{2}\sin 1$

# 例题

例2 证明 $\dfrac{\mathrm{d}\tan x}{\mathrm{d}x} = \sec^2 x$。

证明：$\dfrac{d}{dx}\tan x = \dfrac{d}{dx}\left(\dfrac{\sin x}{\cos x}\right) = \dfrac{(\sin x)'\cos x - \sin x(\cos x)'}{\cos^2 x}$

$= \dfrac{\cos^2 x + \sin^2 x}{\cos^2 x} = \dfrac{1}{\cos^2 x} = \sec^2 x$

同理 $\dfrac{d}{dx}(\cot x) = \dfrac{d}{dx}\left(\dfrac{\cos x}{\sin x}\right) = -\csc^2 x$

# 例题

例3 证明 $\dfrac{\mathrm{d}\sec x}{\mathrm{d}x} = \sec x \tan x$。

证明：$\dfrac{d}{dx}\sec x = \dfrac{d}{dx}\left(\dfrac{1}{\cos x}\right) = \dfrac{-(\cos x)'}{\cos^2 x} = \dfrac{\sin x}{\cos^2 x} = \tan x \cdot \dfrac{1}{\cos x}$

$= \tan x \sec x$

类似 $\dfrac{d}{dx}\csc x = \dfrac{d}{dx}\left(\dfrac{1}{\sin x}\right) = -\csc x \cdot \cot x$

## 求导法则

设$y=f(x)$为$x=\varphi(y)$的反函数，若$x=\varphi(y)$是严格单调的连续函数，在点$y$处可导，且$\varphi'(y)\neq 0$，则$y=f(x)$在对应点$x$处也可导，且

$$f'(x)=\frac{1}{\varphi'(y)}.$$

证明：$\because x=\varphi(y)$严格单调连续 $\therefore y=f(x)$严格单调连续

$\Delta x\to 0\Leftrightarrow \Delta y\to 0$，$\Delta x\neq 0$时 $\Delta y=f(x+\Delta x)-f(x)\neq 0$

$$\therefore f'(x)=\lim_{\Delta x\to 0}\frac{\Delta y}{\Delta x}=\lim_{\Delta y\to 0}\frac{1}{\frac{\Delta x}{\Delta y}}=\frac{1}{\varphi'(y)}$$

# 例题

例4 证明$(a^x)' = a^x \ln a$, 其中$a > 0$ 且$a \neq 1$。

证明：$y = a^x$ $x \in R$ 是 $x = \log_a^y$ $y \in (0, +\infty)$ 的反函数

在$(0, +\infty)$上 $x = \log_a^y$ 严格单调 连续 可导 且

$$\log_a^y = \frac{1}{y \cdot \ln a} \neq 0$$

$$\therefore (a^x)' = \frac{1}{(\log_a^y)'} = y \cdot \ln a = a^x \cdot \ln a$$

特别地 $(e^x)' = e^x$

# 例题

例5 证明$(\arcsin x)' = \dfrac{1}{\sqrt{1-x^2}}$。

证明：$x=\sin y$ 在 $y\in(-\frac{\pi}{2},\frac{\pi}{2})$ 内严格单调、连续可导

且 $(\sin y)' = \cos y \neq 0$

$\therefore (\arcsin x)' = \dfrac{1}{\cos y} = \dfrac{1}{\sqrt{1-\sin^2 y}} = \dfrac{1}{\sqrt{1-x^2}}$ $(-1<x<1)$

类似地 $(\arccos x)' = \dfrac{-1}{\sqrt{1-x^2}}$ $(\arctan x)' = \dfrac{1}{1+x^2}$

$(\operatorname{arccot} x)' = \dfrac{-1}{1+x^2}$

## 链式法则

设$u=\varphi(x)$在点$x_0$处可导，$y=f(u)$在点$u_0=\varphi(x_0)$处可导，则复合函数$y=f(\varphi(x))$在点$x_0$处可导，且有

$$\frac{\mathrm{d}}{\mathrm{d}x}f(\varphi(x))|_{x=x_0}=f'(u_0)\varphi'(x_0)=f'(\varphi(x_0))\varphi'(x_0).$$

$y-u-x$　$\frac{dy}{dx}=\frac{dy}{du}\cdot\frac{du}{dx}$

证明：$y=f(u)$在$u_0$处可导 $\therefore \lim\limits_{u\to u_0}\frac{f(u)-f(u_0)}{u-u_0}=f'(u_0)$

$\therefore u\neq u_0$时 $\frac{f(u)-f(u_0)}{u-u_0}=f'(u_0)+\alpha(u)$ 其中$\alpha(u)$是

(1) $u\to u_0$时无穷小

# 链式法则的证明

$$\Longrightarrow f(u)-f(u_0)=f'(u_0)\cdot(u-u_0)+\alpha(u)\cdot(u-u_0) \quad (2)$$

当$u=u_0$时 规定 $\alpha(u_0)=0$ 则$\alpha(u)$在$u_0$处连续

将$u=\varphi(x)$代入(2)得

$$\frac{f(\varphi(x))-f(\varphi(x_0))}{x-x_0}=f'(u_0)\frac{\varphi(x)-\varphi(x_0)}{x-x_0}+\alpha(\varphi(x))\frac{\varphi(x)-\varphi(x_0)}{x-x_0}$$

两边取极限 $x\to x_0$ 得

$$\frac{d}{dx}f(\varphi(x))\Big|_{x=x_0}=f'(u_0)\varphi'(x_0)+0\cdot\varphi'(x_0)=f'(u_0)\cdot\varphi'(x_0)$$

$$\big(f(\varphi(x))\big)'=f'(u)\cdot\varphi'(x)=f'(\varphi(x))\cdot\varphi'(x)$$

$(f(\varphi(x)))'$是对$x$求导，$f'(\varphi(x))$是$f(u)$在$u=\varphi(x)$处的导数 $f'(u)\big|_{u=\varphi(x)}$

# 例题

例1 设$y=\ln(x+\sqrt{x^2+1})$,求$y'$。

解: $y=\ln(x+\sqrt{x^2+1})$是 $y=\ln u$ 与 $u=x+\sqrt{x^2+1}$ 的复合

$$\therefore \frac{dy}{dx}=\frac{dy}{du}\cdot\frac{du}{dx}=\frac{1}{u}\cdot(x+\sqrt{x^2+1})'$$

$$=\frac{1}{x+\sqrt{x^2+1}}\cdot\left(1+\frac{2x}{2\sqrt{x^2+1}}\right)$$

$$=\frac{1}{\sqrt{x^2+1}}$$

$y$
|
$u$
|
$x$

($z=\sqrt{x^2+1}$ 可视为 $z=v^{\frac{1}{2}}$ 与 $v=x^2+1$ 的复合)

$$\frac{dz}{dv}=\frac{1}{2}v^{-\frac{1}{2}},\quad \frac{dv}{dx}=2x$$

## 可推广到两个以上函数的复合

$z = f(y), y = g(x), x = h(t)$三个函数复合得$z = f(g(h(t)))$, 则求导公式为

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\mathrm{d}z}{\mathrm{d}y} \cdot \frac{\mathrm{d}y}{\mathrm{d}x} \cdot \frac{\mathrm{d}x}{\mathrm{d}t}.$$

z-y-x-t

# 例题

例2 设$y=\tan^2\frac{1}{x}$,求$y'$。

解. $y=\tan^2\frac{1}{x}$ 是 $y=u^2$, $u=\tan v$, $v=\frac{1}{x}$ 复合而成

$$\therefore \frac{dy}{dx}=\frac{dy}{du}\cdot\frac{du}{dv}\cdot\frac{dv}{dx}$$

$$=2u\cdot\sec^2 v\cdot\left(-\frac{1}{x^2}\right)$$

$$=2\tan\frac{1}{x}\cdot\sec^2\frac{1}{x}\cdot\left(-\frac{1}{x^2}\right)$$

# 例题

例3 设$y=\dfrac{2+\sin^2 x}{\cos 2x}$,求$y'$。

# 例题

例4 证明$(\frac{1}{f^n(x)})' = \frac{-nf'(x)}{f^{n+1}(x)}$, 并设$y = \frac{1}{\arctan^2 x}$,求$y'$。

# 例题

例5 求幂指函数$y=(u(x))^{v(x)}$ 的导数，其中$u(x)>0, v(x)$均可导。

# 例题

例6 求$y = \ln|x|$的导数。

# 例题

例7 设$f(x)=\begin{cases}1+3x & x<0\\ e^{2x} & x\geq 0\end{cases}$,求$f'(x)$。

# 例题

例8 设$y = f(1 + f(f(x)))$其中$f(x)$为可导函数,求$y'$。

## 定义

若函数$f(x)$的导函数$f'(x)$在点$x_0$处可导,则称$f'(x)$在$x_0$点的导数为$f(x)$在$x_0$点的二阶导数.记作$f''(x_0)$,即

$$f''(x_0)=\lim_{x\to x_0}\frac{f'(x)-f'(x_0)}{x-x_0}.$$

二阶及二阶以上的导数统称为高阶导数.

函数$y=f(x)$在$x_0$点的$n$阶导数记作

$$f^{(n)}(x_0),\quad \frac{\mathrm{d}^n f(x)}{\mathrm{d}x^n}\Big|_{x=x_0},\quad y^{(n)}\Big|_{x=x_0},\quad \frac{\mathrm{d}^n y}{\mathrm{d}x^n}\Big|_{x=x_0}.$$

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## 例1

设$y = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$, 求$y^{(n)}$.

解: $y' = a_1 + 2a_2 x + 3a_3 x^2 + \cdots + n a_n x^{n-1}$,

$y'' = 2a_2 + 3 \cdot 2a_3 x + \cdots + n(n-1) a_n x^{n-2}$,

依次类似可得$y^{(n)} = n! a_n$. 且$y^{(n+1)} = y^{(n+2)} = \cdots = 0$.

## 例2

设$y = e^{ax}$,求$y^{(n)}$.

解: $y' = a e^{ax}$, $y'' = a^2 e^{ax}$, $\cdots$, $y^{(n)} = a^n e^{ax}$.

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依次类似可得$y^{(n)} = n! a_n$. 且$y^{(n+1)} = y^{(n+2)} = \cdots = 0$.

## 例2

设$y = e^{ax}$,求$y^{(n)}$.

解: $y' = a e^{ax}$, $y'' = a^2 e^{ax}$, $\cdots$, $y^{(n)} = a^n e^{ax}$.

特别地: $(e^x)^{(n)} = e^x$.

## 例1

设$y = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$, 求$y^{(n)}$.

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## 例3

设$y = \sin x$, 求$y^{(n)}$.

解：$y' = \cos x = \sin(x + \frac{\pi}{2})$

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设$y = \ln\sin x$, 求$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2}$.

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## 高阶导数的运算法则

1. $\frac{\mathrm{d}^n}{\mathrm{d}x^n}[f(x)\pm g(x)]=\frac{\mathrm{d}^n}{\mathrm{d}x^n}f(x)\pm\frac{\mathrm{d}^n}{\mathrm{d}x^n}g(x).$

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设$y = x^2e^x$, 求$y^{(50)}$.

解:

$$\begin{aligned} y^{(50)} &= \sum_{k=0}^{50} C_{50}^k (x^2)^{(k)} (e^x)^{(50-k)}, \\ &= C_{50}^0 x^2 e^x + C_{50}^1 2x e^x + C_{50}^2 2e^x \\ &= (x^2 + 100x + \frac{50 \cdot 49}{2} \cdot 2)e^x. \end{aligned}$$

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解:

$$\begin{aligned} y^{(50)} &= \sum_{k=0}^{50} C_{50}^k (x^2)^{(k)} (e^x)^{(50-k)}, \\ &= C_{50}^0 x^2 e^x + C_{50}^1 2x e^x + C_{50}^2 2e^x \\ &= (x^2 + 100x + \frac{50 \cdot 49}{2} \cdot 2)e^x. \end{aligned}$$

## 例5

设$y = x^2e^x$, 求$y^{(50)}$.

解:

$$\begin{aligned} y^{(50)} &= \sum_{k=0}^{50} C_{50}^k (x^2)^{(k)} (e^x)^{(50-k)}, \\ &= C_{50}^0 x^2 e^x + C_{50}^1 2x e^x + C_{50}^2 2e^x \\ &= (x^2 + 100x + \frac{50 \cdot 49}{2} \cdot 2)e^x. \end{aligned}$$

## 例5

设$y = x^2e^x$, 求$y^{(50)}$.

解:

$$\begin{aligned} y^{(50)} &= \sum_{k=0}^{50} C_{50}^k (x^2)^{(k)} (e^x)^{(50-k)}, \\ &= C_{50}^0 x^2 e^x + C_{50}^1 2x e^x + C_{50}^2 2e^x \\ &= (x^2 + 100x + \frac{50 \cdot 49}{2} \cdot 2)e^x. \end{aligned}$$

# 例题

例6 设$y = f(f(x))$,求$y'$.

# 例题

例7 试从$\dfrac{\mathrm{d}x}{\mathrm{d}y}=\dfrac{1}{y'}$导出$\dfrac{\mathrm{d}^2x}{\mathrm{d}y^2}=-\dfrac{y''}{(y')^3}$.

# 例题

例8 设$y = \ln(3 + 7x - 6x^2)$的$n$阶导数.